Yarkowsky´s effect simulation

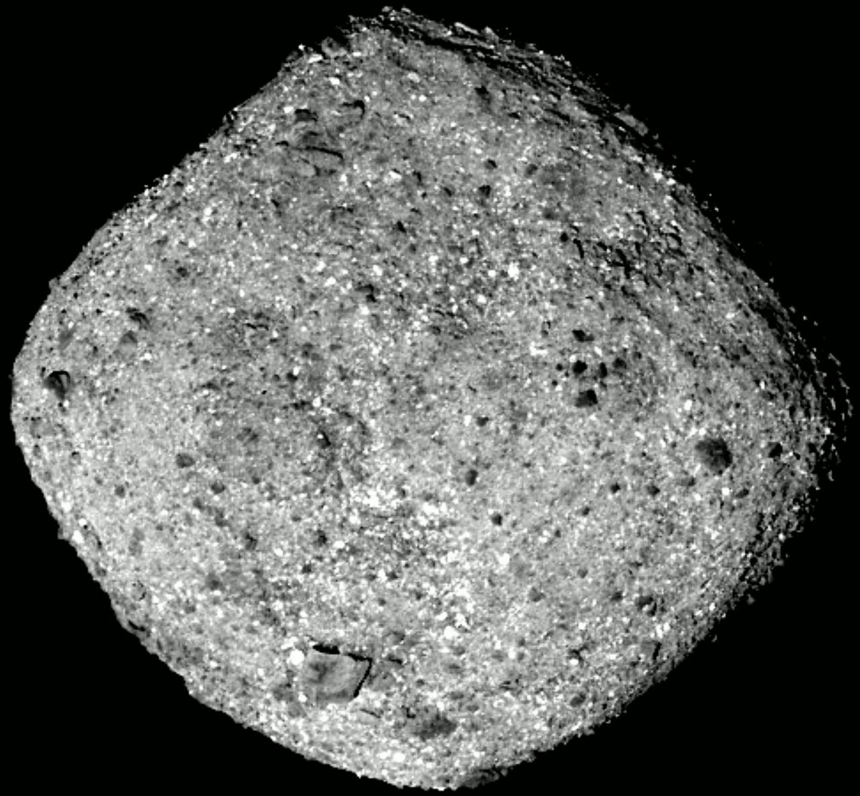
- Final presentation -

Image: NASA

DSSC Core Course, June 2020
Selver Pepić

Remember Chelyabinsk 2013?

Earth, meet Bennu ☺

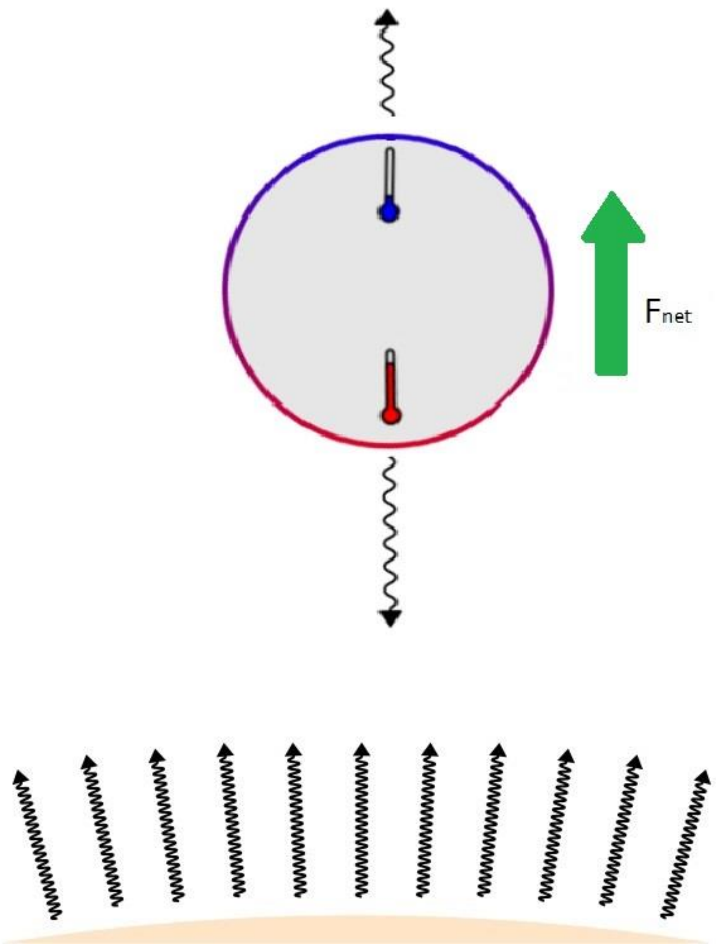


- Diameter: ~**450 m**
  (Chelyabinsk ~ 20 m, Dinosaur killer ~ 50 km)
- Mass: ~**7.33 × 10^{10} kg**
- Energy: **1.20 × 10^{6} kton TNT**
  80000 x Hiroshima bomb
- Aphelion: **~1.36 AU**
- Perihelion: ~**0.90 AU (!)**

- „Cumulative 1-in-2700 chance of impacting the Earth between 2175 and 2199“ [NASA cneos]
- „The analysis of impact probabilities ... is strongly dependent on the **Yarkowsky effect**“ [Milani et al. 2009]

Yarkowsky´s effect = asymmetric thermal recoil

Image by: Graevemoore, Wikipedia



Asymmetric temperature distribution
→ asymmetric thermal radiation
→ nonzero **net recoil force**
(photons carry momentum!)

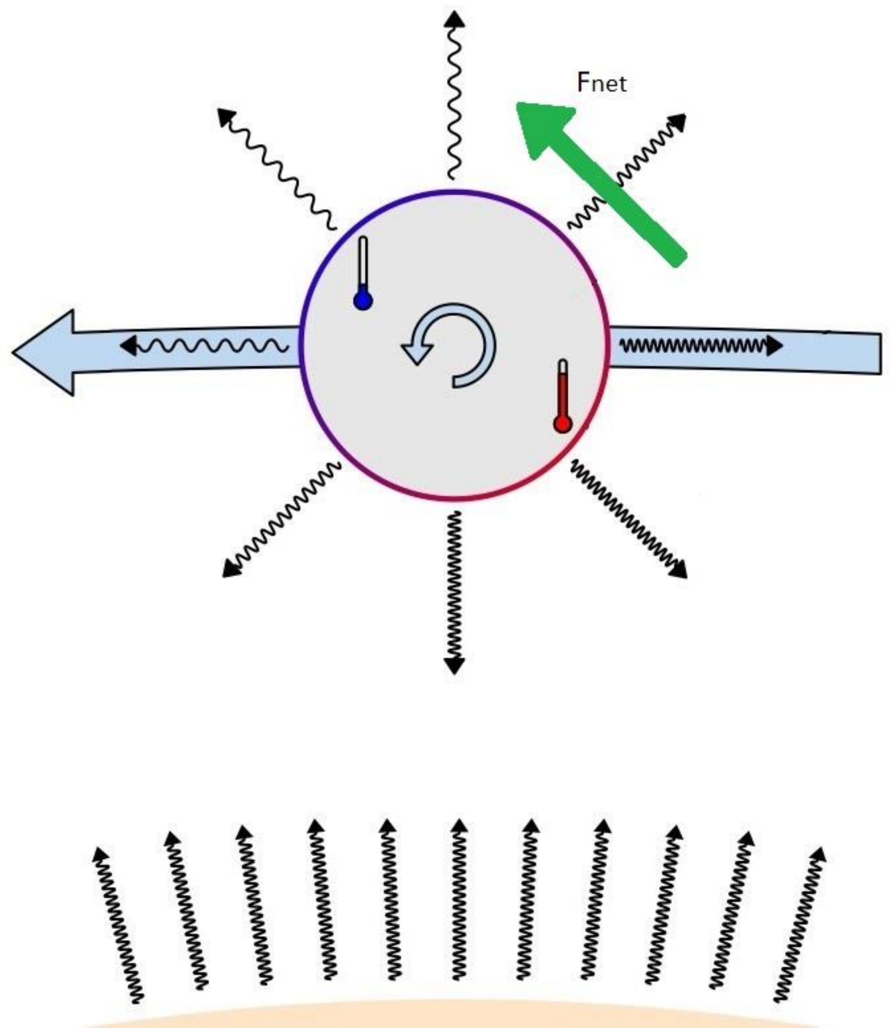
Non-rotating asteroid illuminated by the Sun:

- asymmetric temperature
- net thermal recoil away from the Sun
- effectively same as weaker gravitational pull

→ **no change** in energy or angular momentum
→ asteroid´s orbit becomes slighly more elliptical, but the orbit is stable

Yarkowsky´s effect = asymmetric thermal recoil

Image by: Graevemoore, Wikipedia



Rotating asteroid:

- „day and night“ cycle on the surface
- due to finite **thermal conductivity** („inertia“) hottest/coldest points **not aligned with the Sun**
- net thermal recoil has a **tangential component**

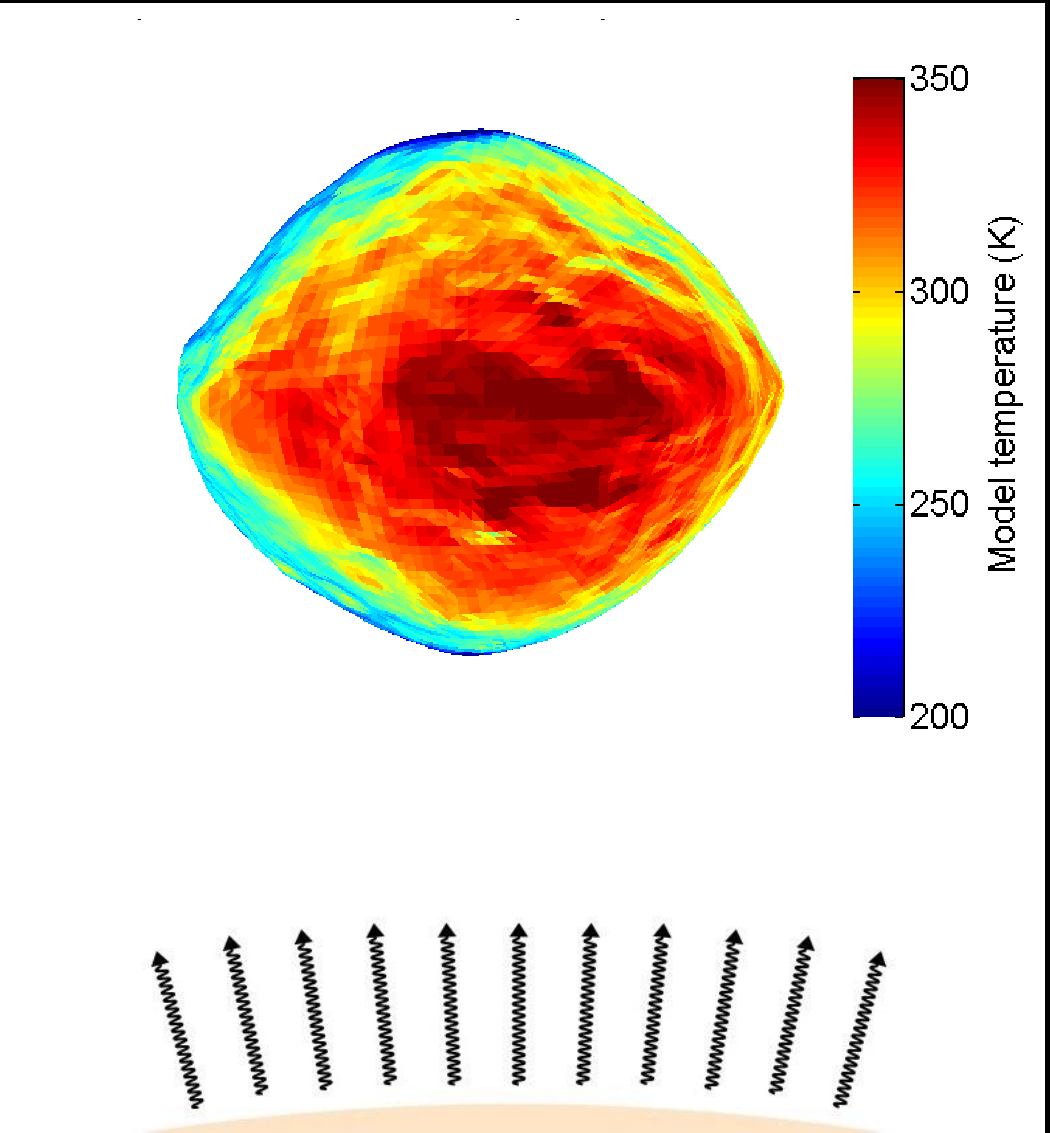
→ energy and angular momentum will change !

→ asteroid´s orbit unstable (spiral in/out)

→ makes **prediction of motion harder**

→ can push asteroids from the asteroid belt towards Earth, thus increasing probability of an impact

Yarkowsky´s effect = asymmetric thermal recoil



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Goals

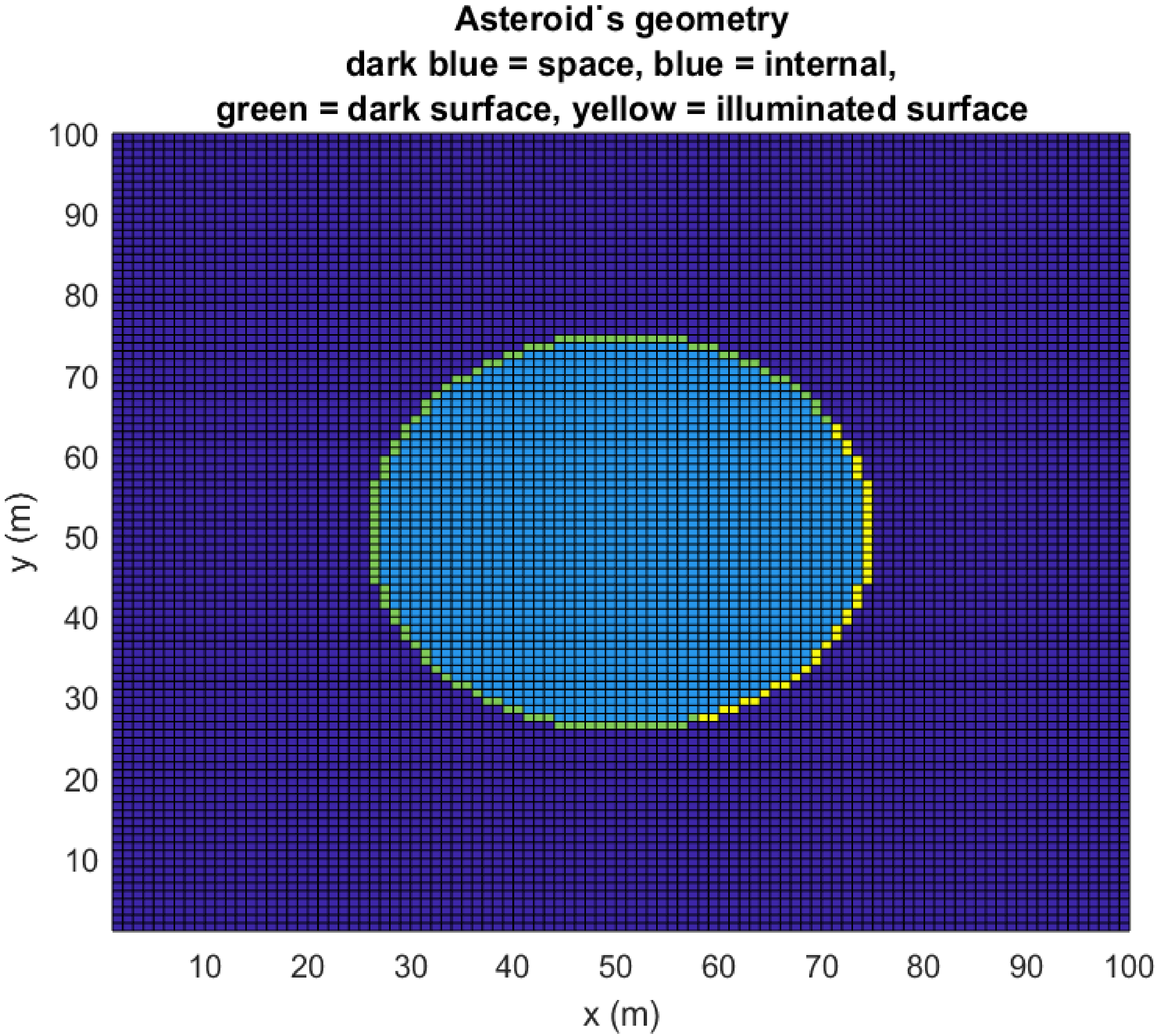
1. Simulate Yarkowsky´s effect for a rotating symmetrical asteroid

2. Simulate orbit of an asteroid subject to combined forces of Sun´s gravity and Yarkowsky´s effect

Geometry

- **Square** asteroid used for testing the code
- **Spherical** asteroid used for final results

- **Discretization:** rectangular grid



Physics: heat diffusion

- Temperature gradient leads to a **heat flow**:

$$\vec{J} = -k\,\vec{\nabla}T \approx -k\,(T_{i+1} - T_i)/\Delta x$$

- (Net) Heat flow into a „cell“ leads to an increase in the temperature:

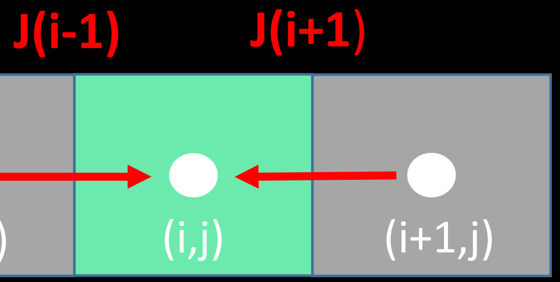
$$\rho c_v \frac{\partial T}{\partial t} = -\vec{\nabla} \cdot \vec{J}$$

- **Discretization:** Explicit forward Euler method (1D)

$$\frac{\rho c_v}{\Delta t}\left(T_i^{k+1} - T_i^k\right) = \frac{k}{(\Delta x)^2}\left(T_{i-1}^k - T_i^k\right) + \frac{k}{(\Delta x)^2}\left(T_{i+1}^k - T_i^k\right)$$

leading to:

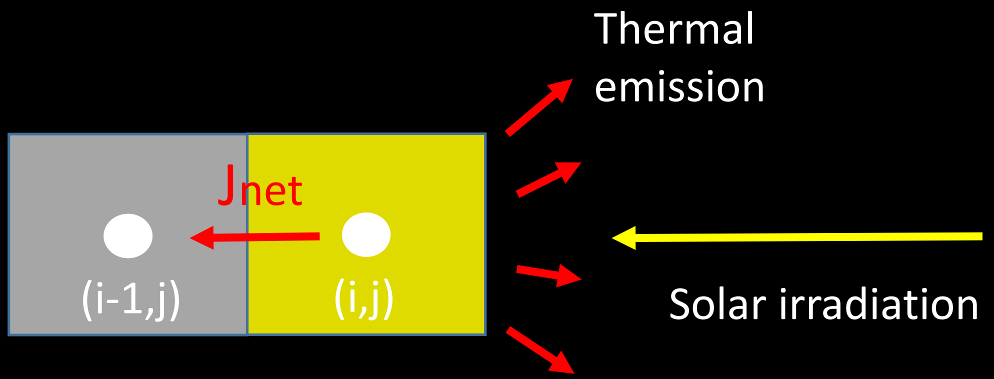
$$T_i^{k+1} = \left(1 - 2\frac{k}{\rho c_v}\frac{\Delta t}{(\Delta x)^2}\right) T_i^k + \frac{k}{\rho c_v}\frac{\Delta t}{(\Delta x)^2}\left(T_{i-1}^k + T_{i+1}^k\right)$$



Physics: energy in/out-flow (boundary conditions)

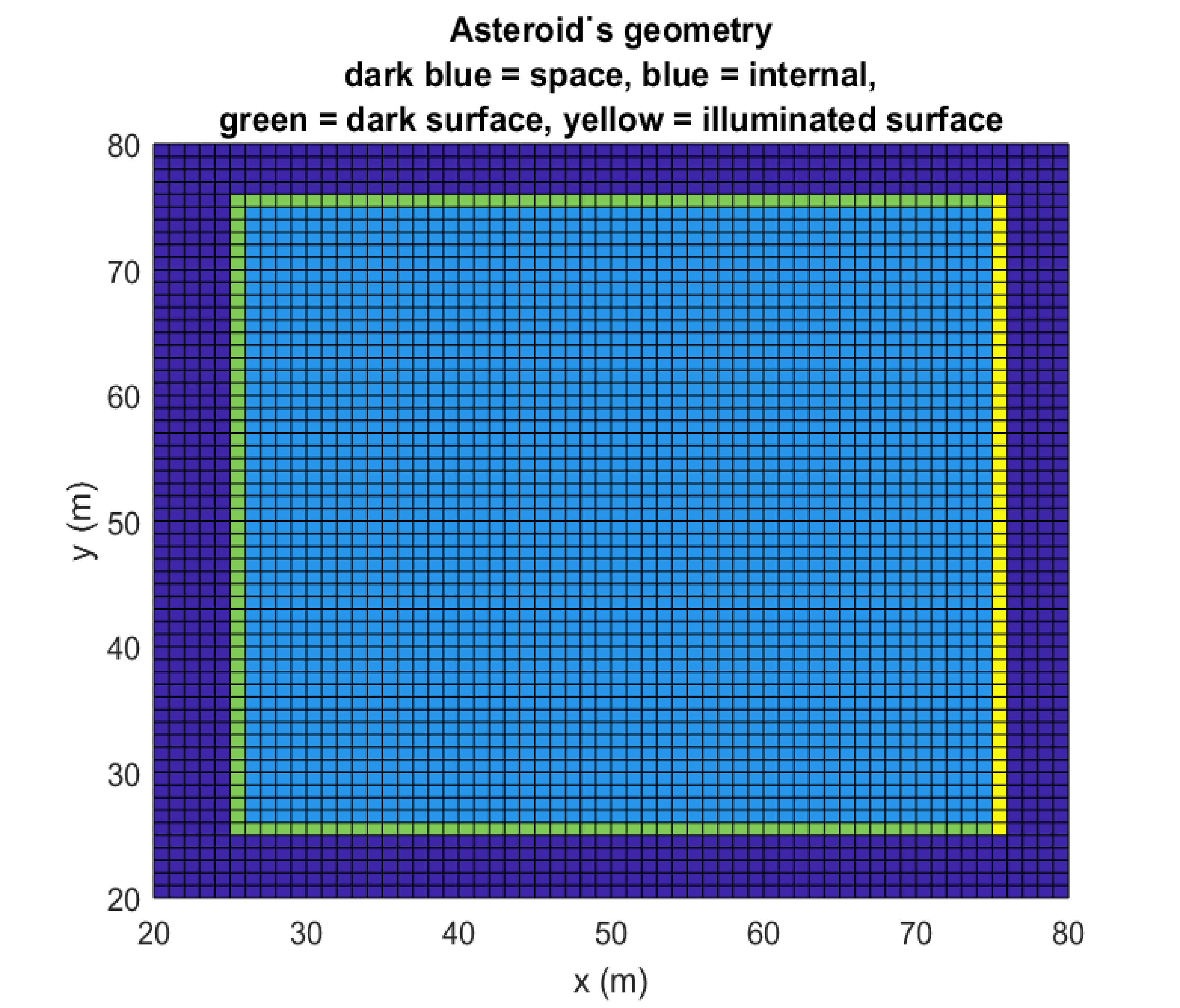
- Boundary conditions at the surface:
  - **Heat inflow** due to the solar irradiation $\boldsymbol{I_o}$
  - **Heat outflow** due to the thermal emission $\mathrm{I_{out}} = \boldsymbol{\sigma T^4}$
- Both conditions specify the temperature gradient $-\boldsymbol{k}\,\boldsymbol{\nabla T} = \boldsymbol{J_{net}}$ at the surface.
- We set the surface node temperature to the correct temperature gradient (relative to its neighbor):

$$\mathrm{T_i} = \mathrm{T_{i-1}} + \left(I_o - \sigma T_i^4\right)/k$$

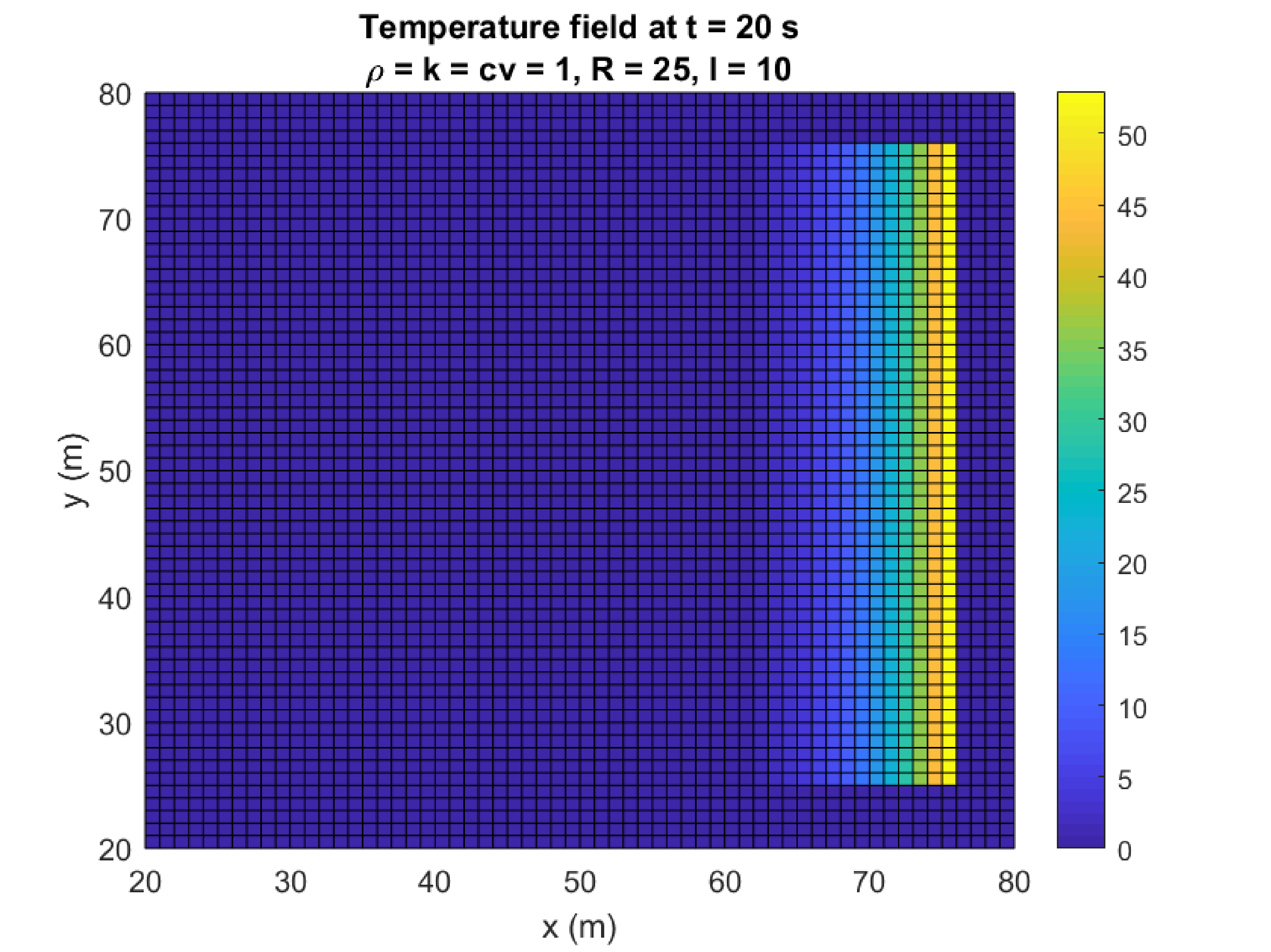


Sanity check: square asteroid

- **Note:** Low values of physical variables used for simplicity and faster convergence



Sanity check: square asteroid



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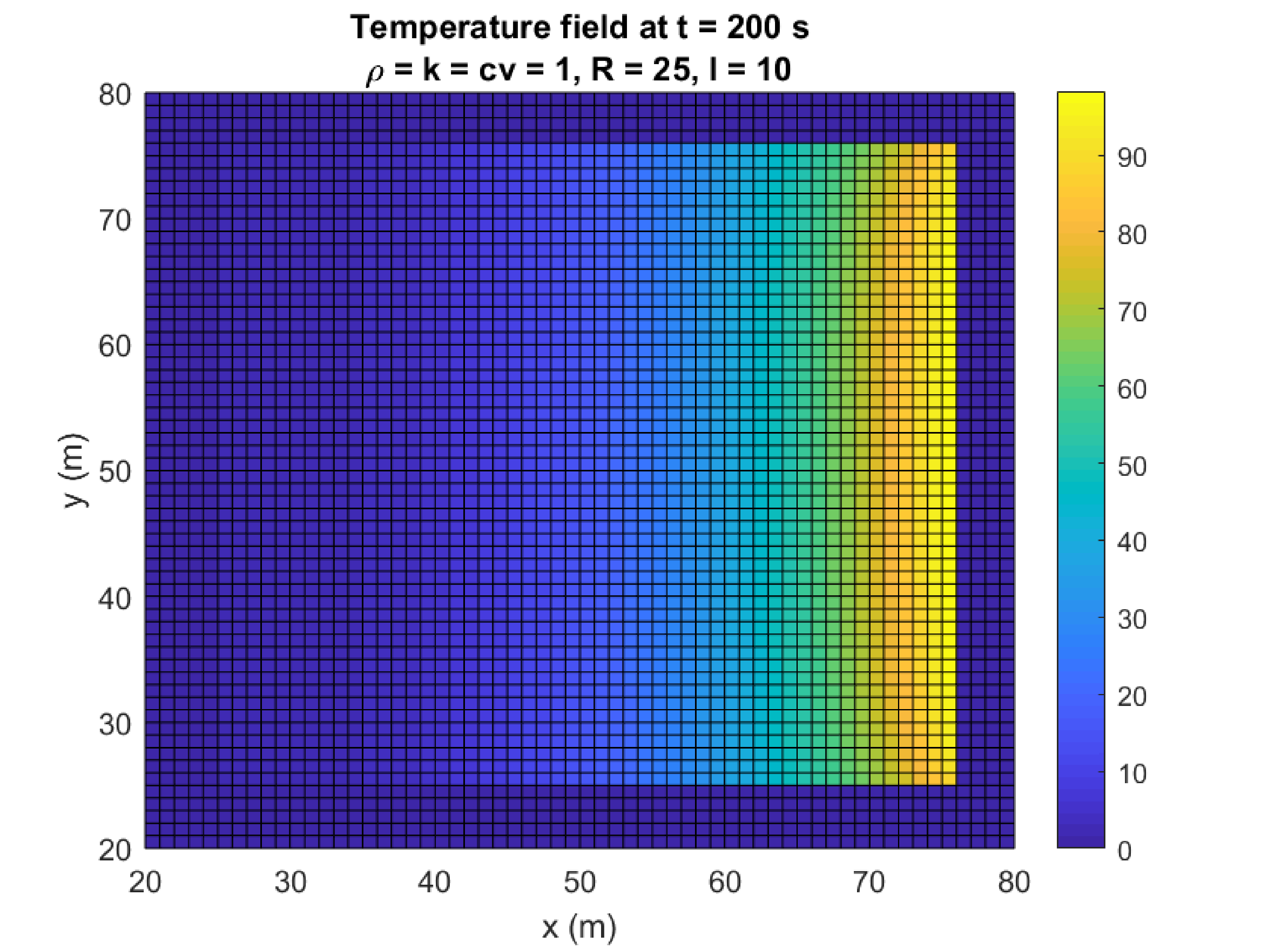
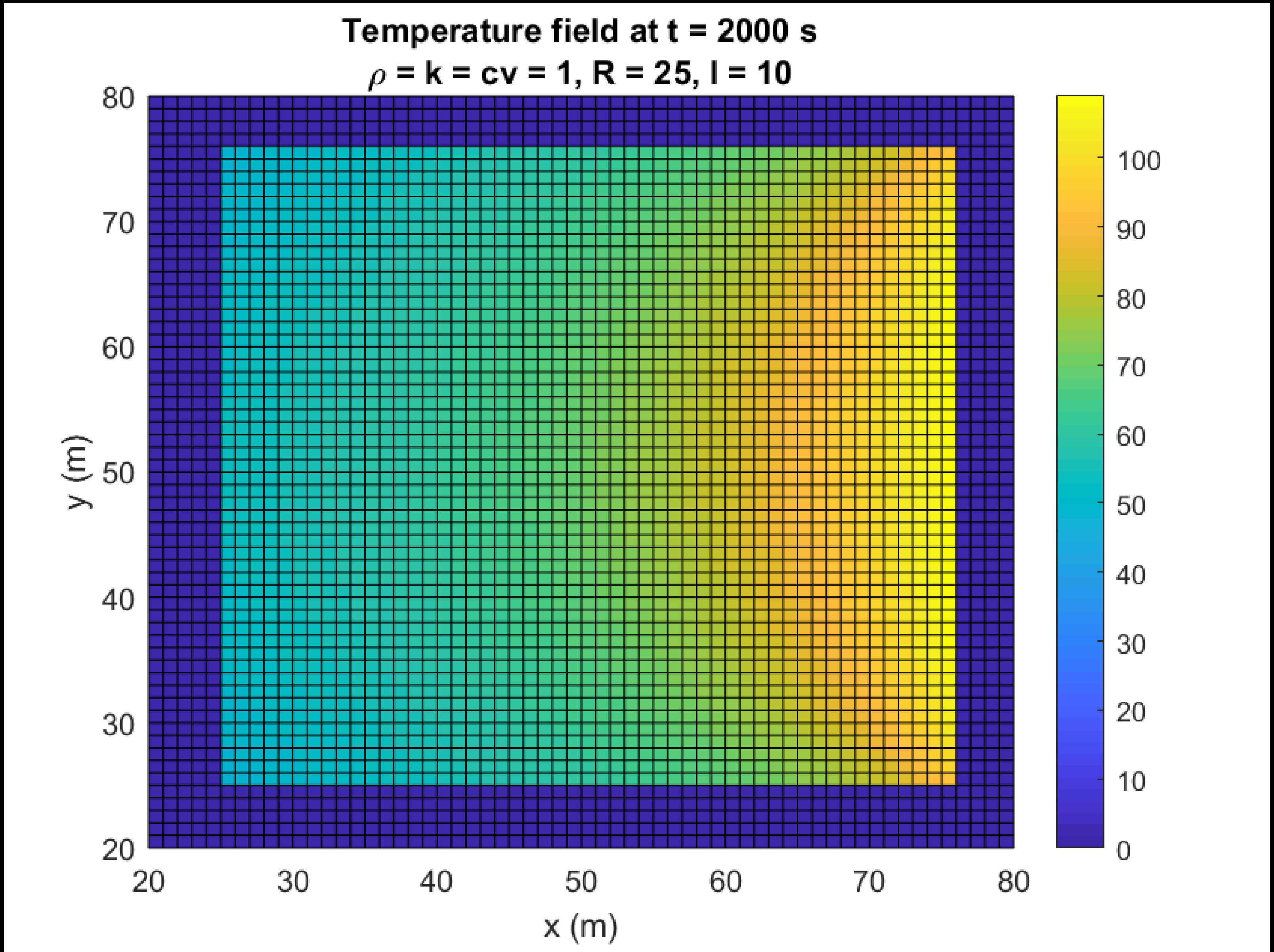


Sanity check: square asteroid

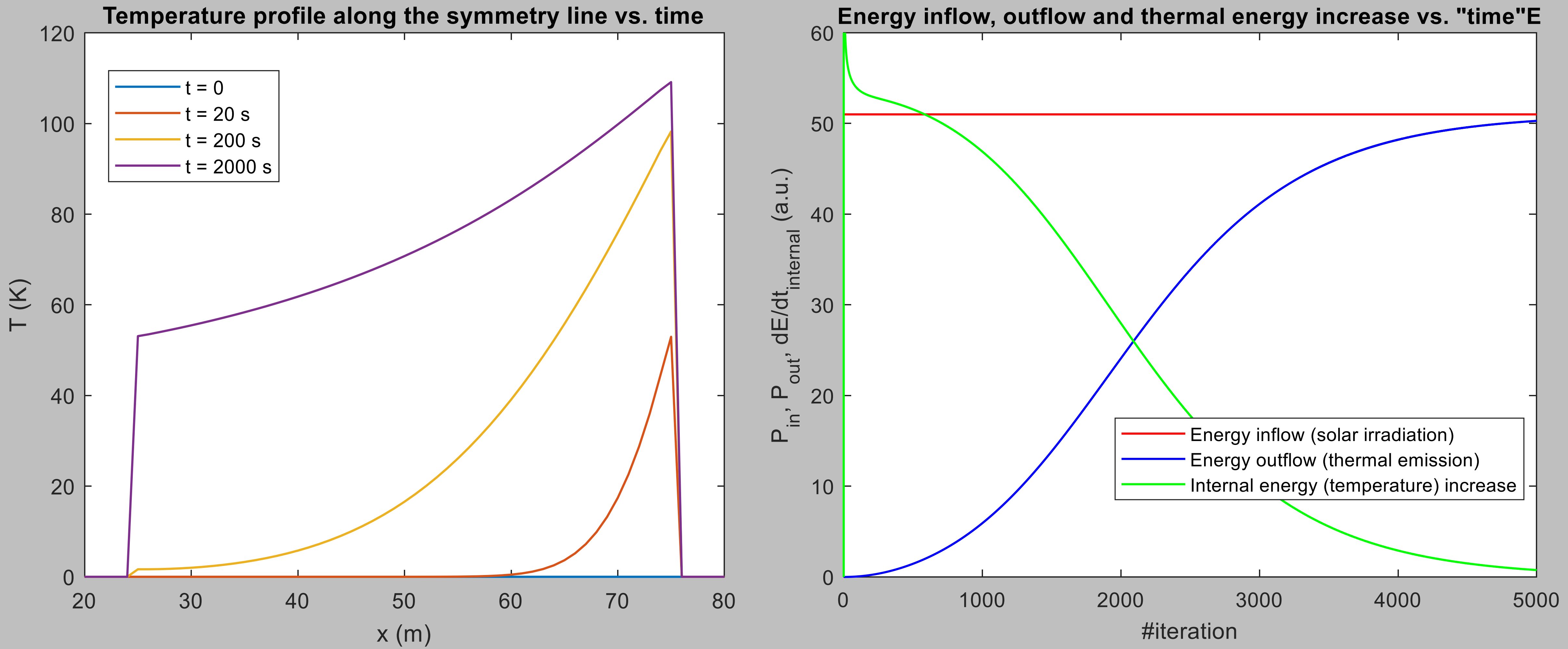
- Theoretical final average temperature: **81 K**
- Simulation final average temperature: **71 K**

- **Yarkowsky accelerations:**
  - a_x = -0.17 a.u.
  - a_y = 1.0 x 10^{-9} a.u.

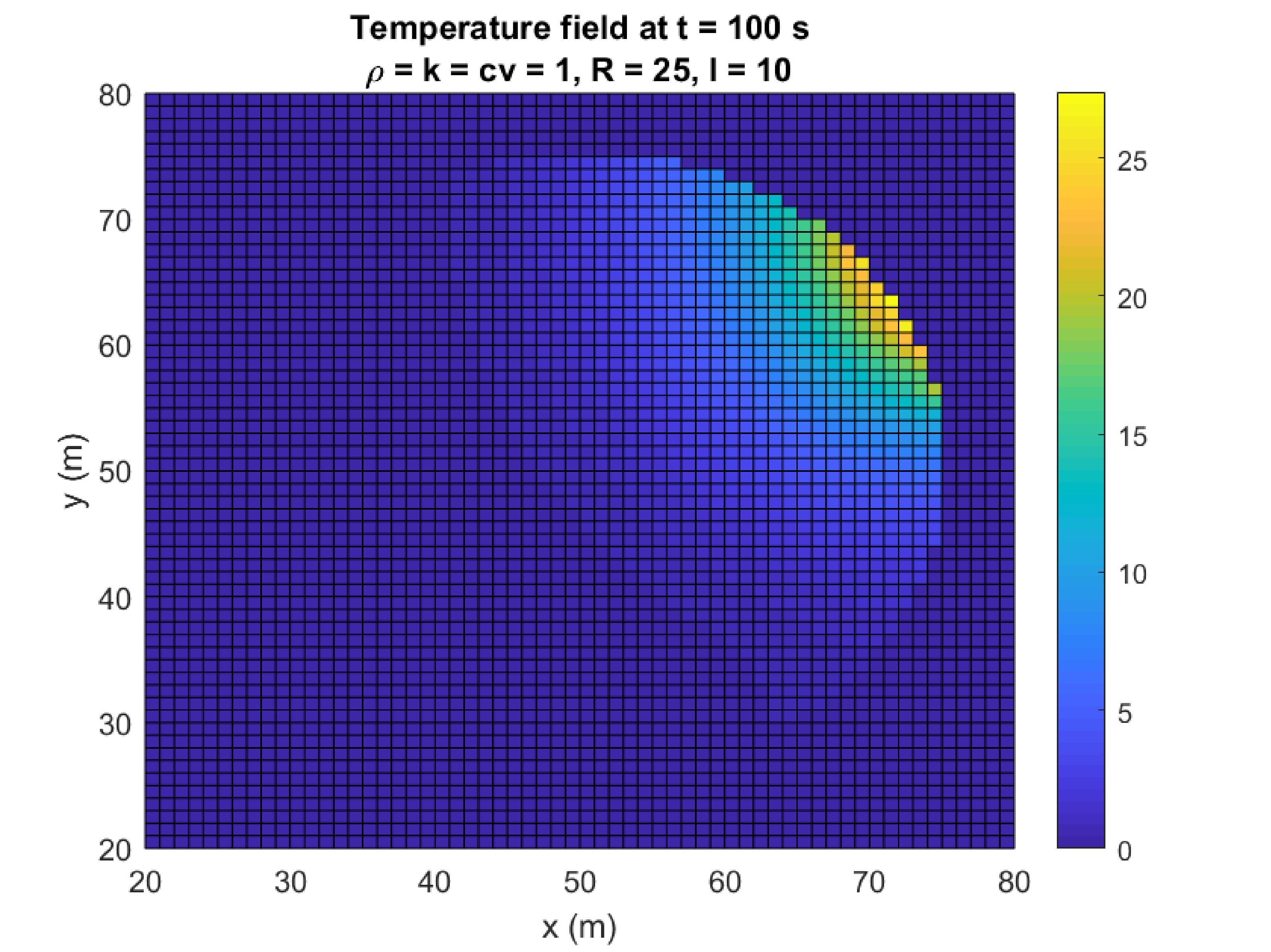
  - Note: values are not-so-informative since all parameters were normalized.



Energy in/out flow and thermal energy increase are balanced



Spherical asteroid (w = 0)



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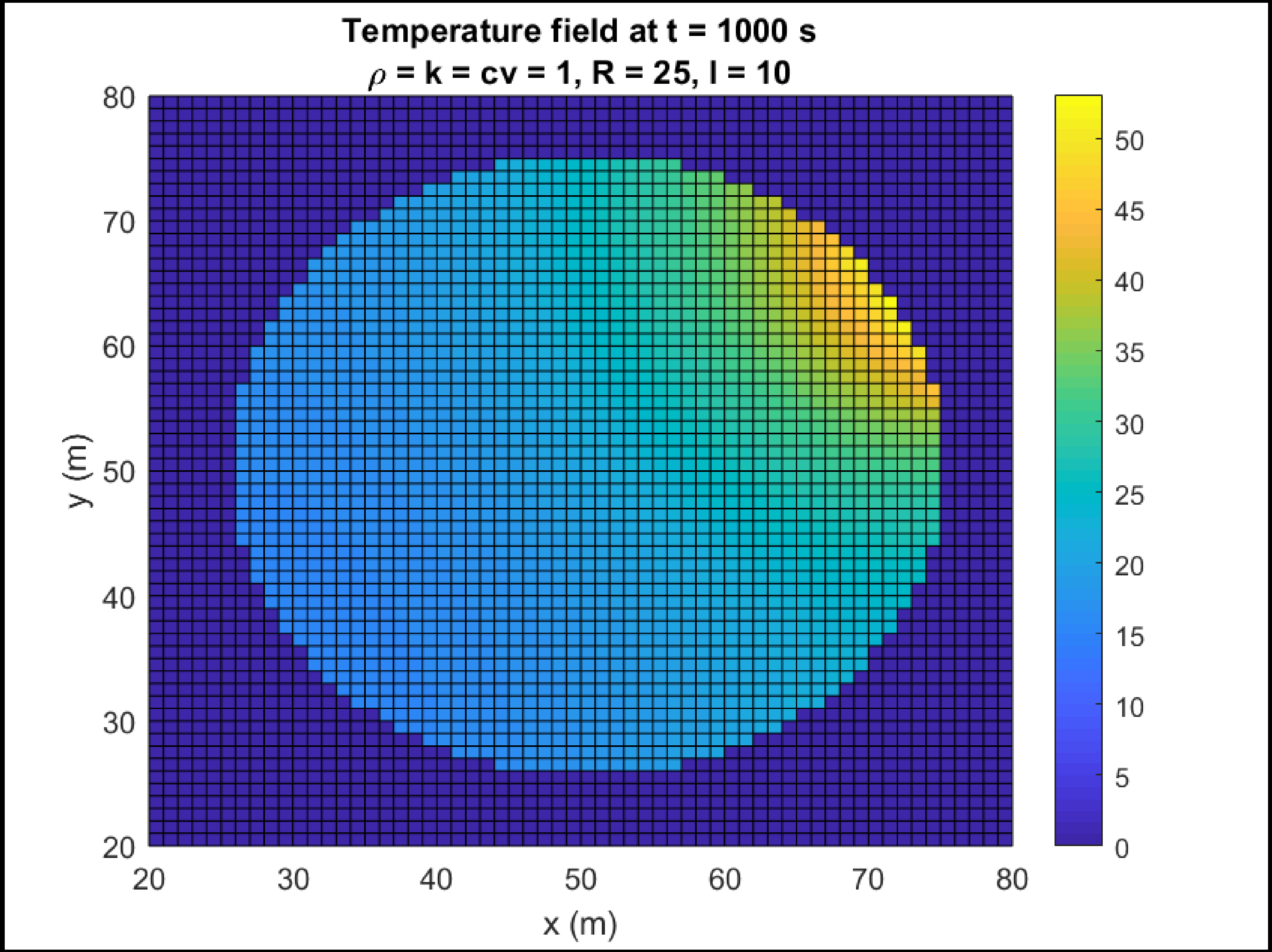
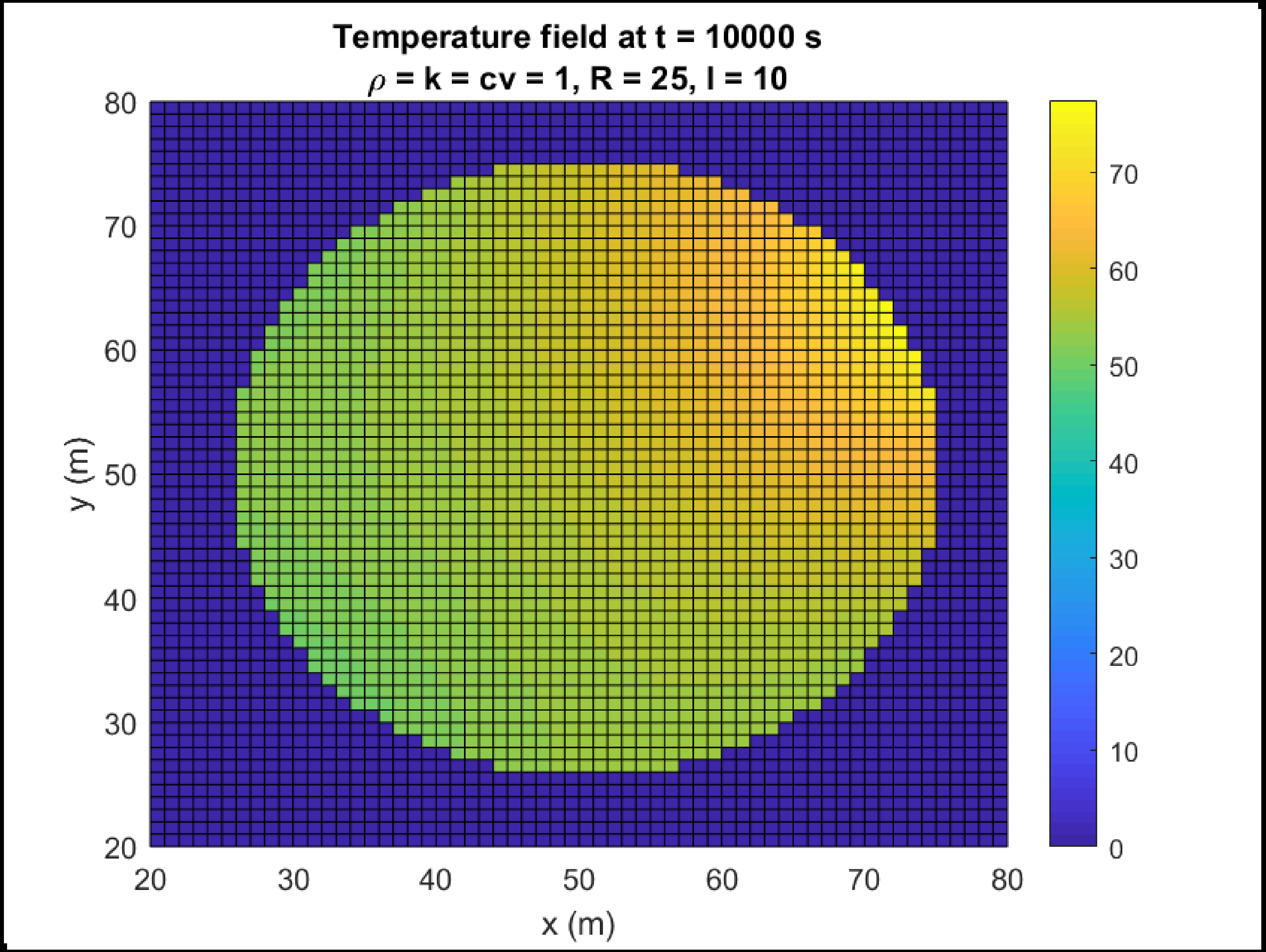


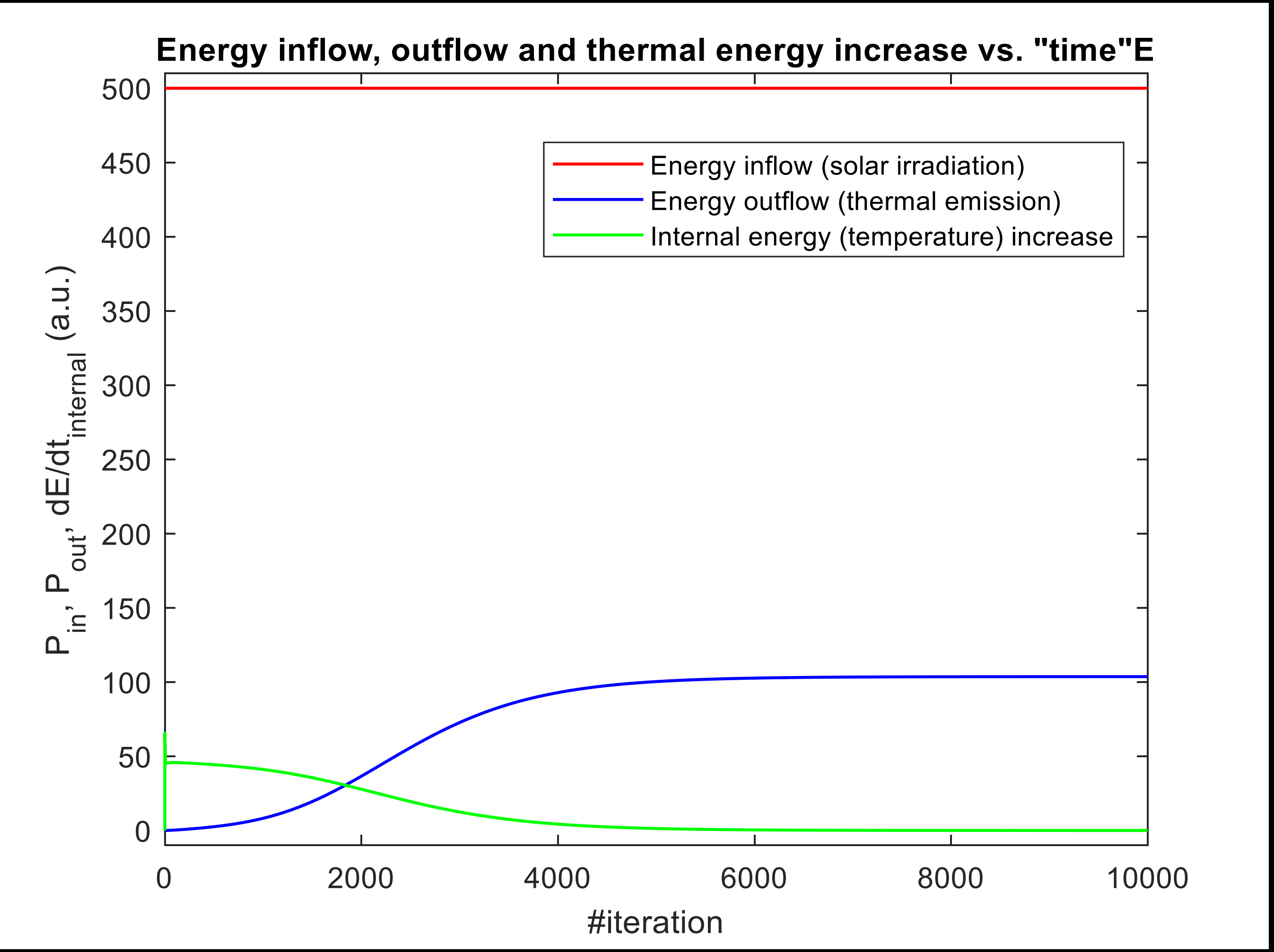
Spherical asteroid (w = 0)

- Theoretical final average temperature: **81 K**
- Simulation final average temperature: **57 K**

- **Cause of discrepancy?**

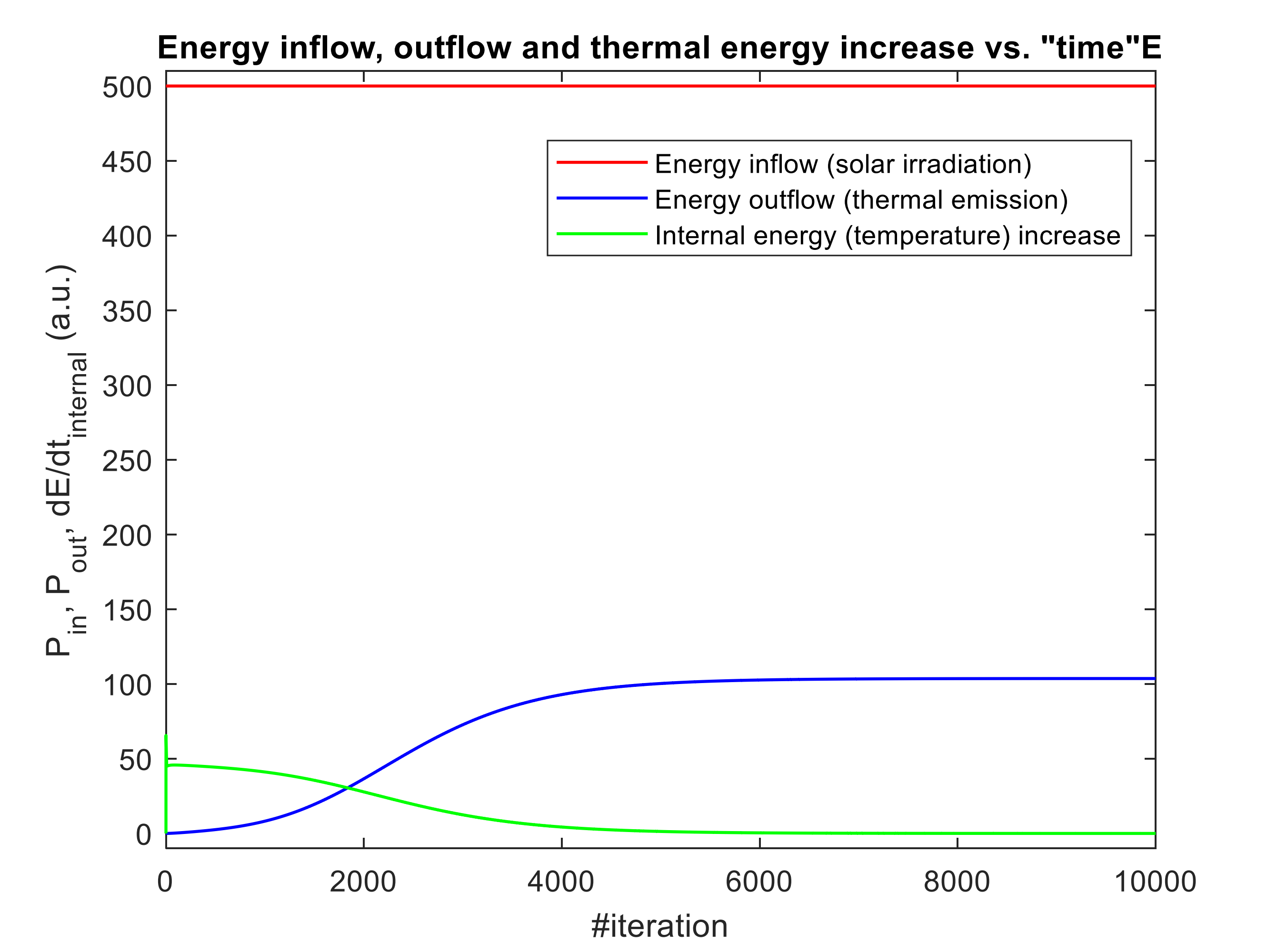


Energy in/out flow and thermal energy increase **NOT** balanced



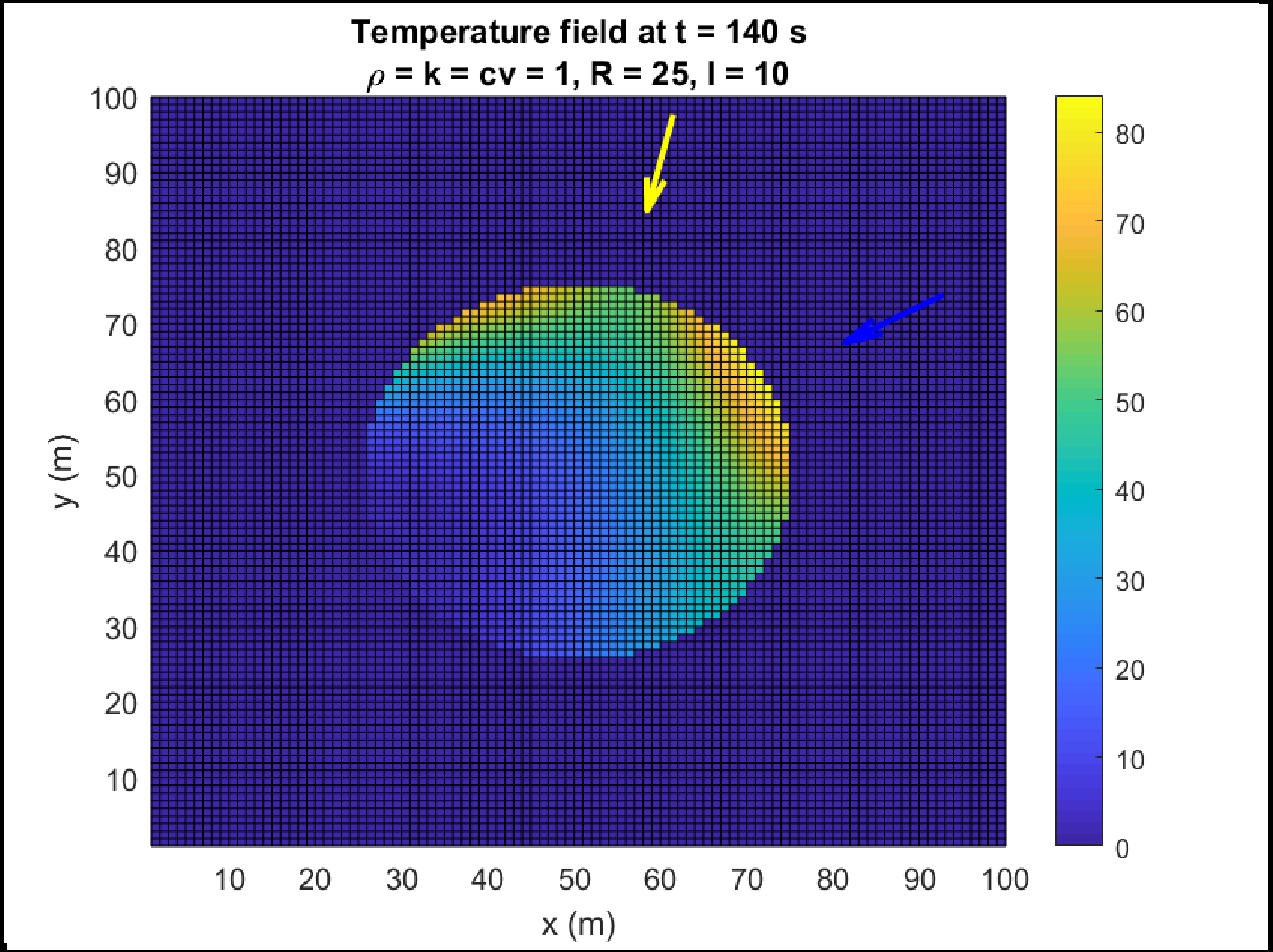
- Major discrepancy between in/out energies.
- Problem persists for higher temporal/spatial resolution.
- Situation better if emission is „disabled“, but discrepancy still present.
- Cause of this error is still unknown (for now).

Energy in/out flow and thermal energy increase **NOT** balanced

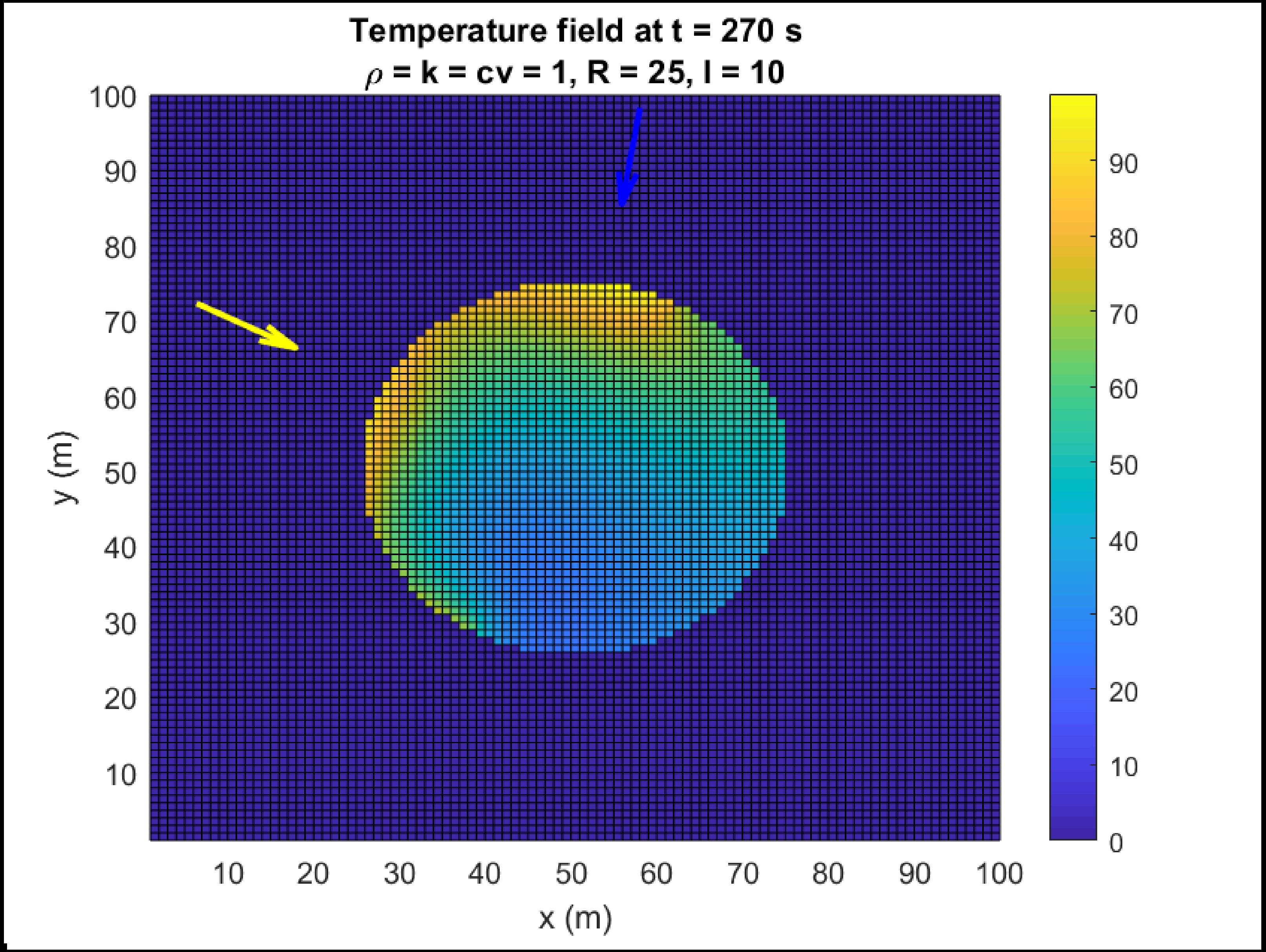


- This prevents us from getting proper results at this stage
- ...but just for fun, here are the temperature plots for a rotating asteroid ☺

Spherical asteroid (rotating))



Spherical asteroid (rotating)



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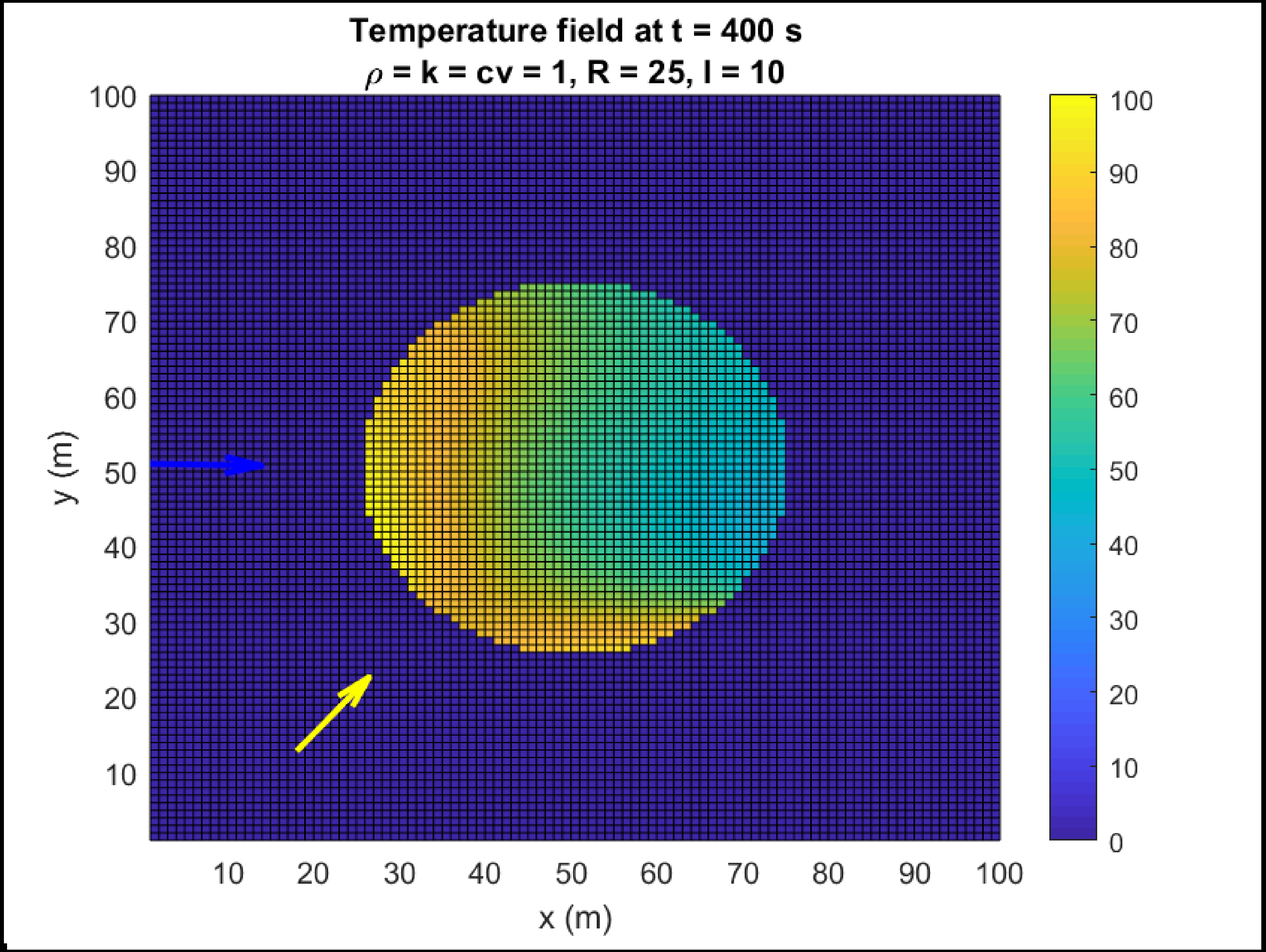

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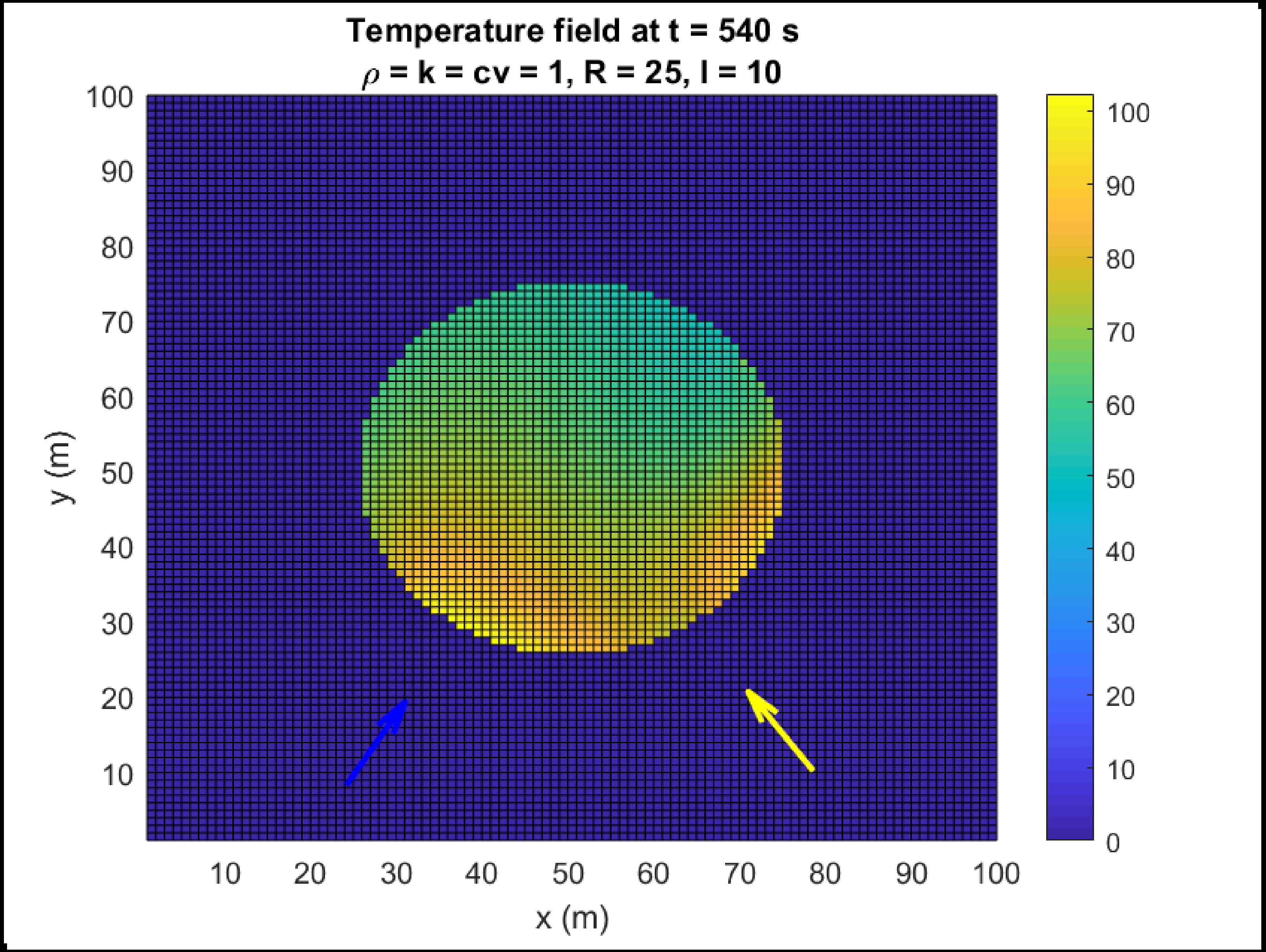
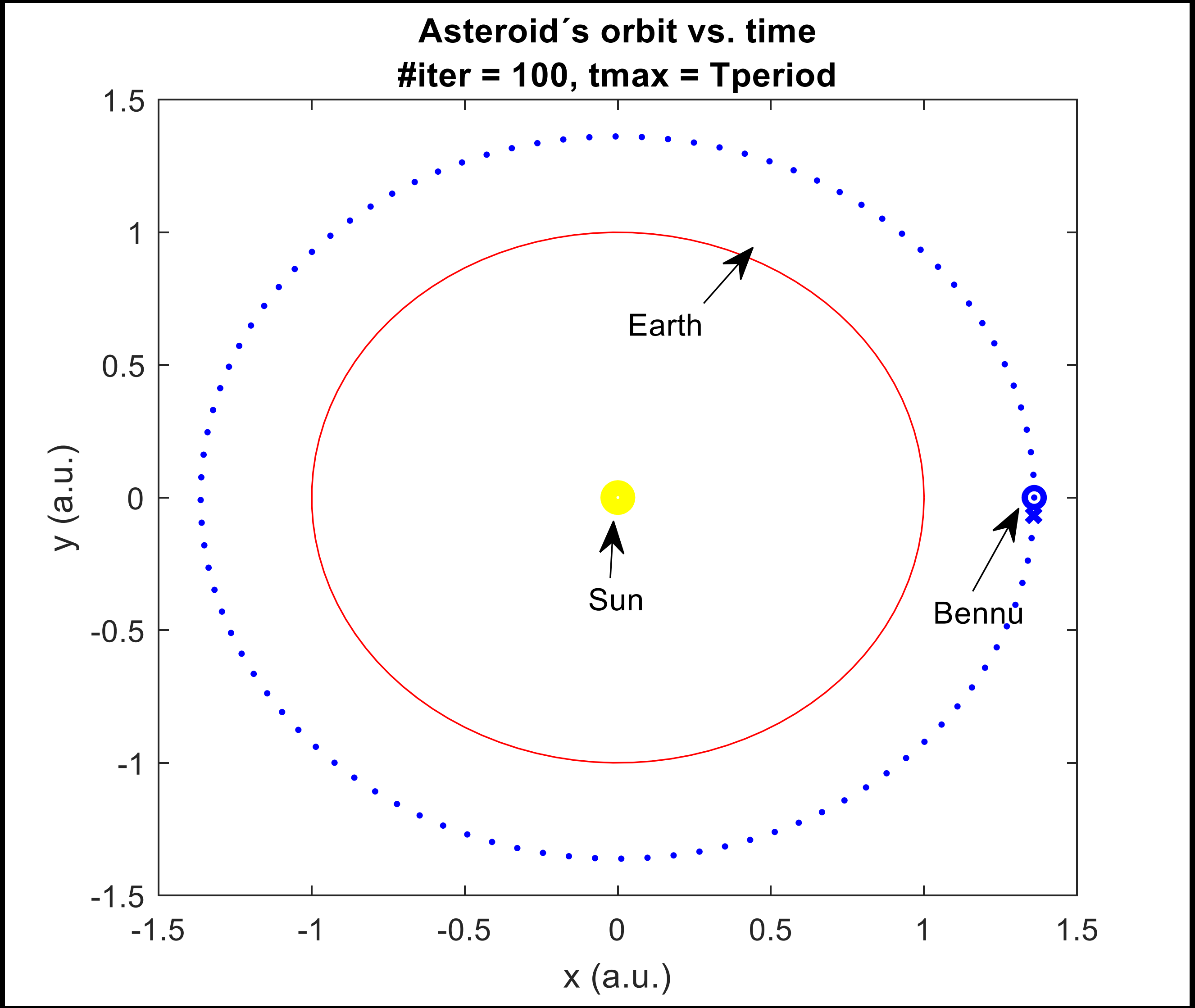


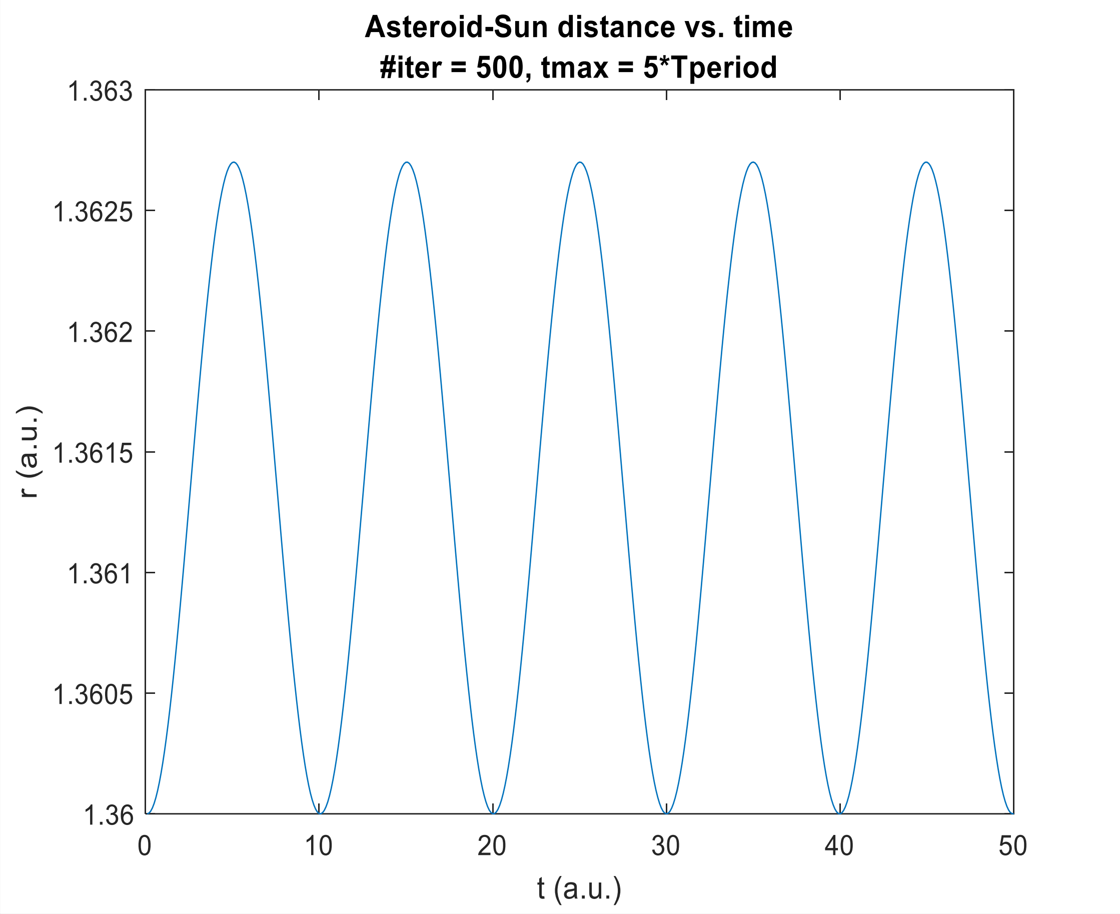
Orbital simulation

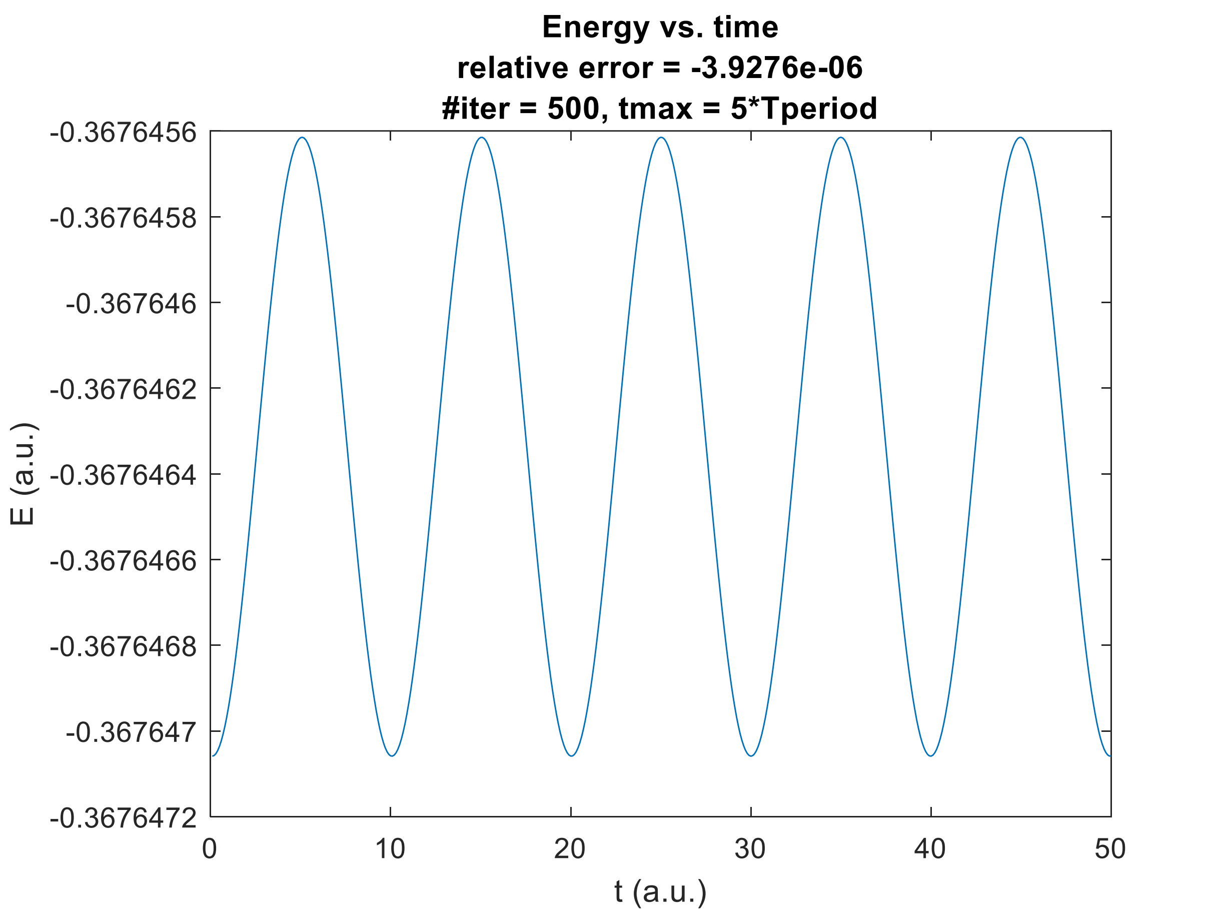
- **Goal:** simulate motion of an asteroid influenced by both the gravity and the Yarkowsky effect

- To avoid numerical energy loss/gain, we used a simplectic integrator (Verlet algorithm)
  → guarantees energy conservation
  → direct discretisation of the 2nd Newton´s law ($d^2x/dt^2 = F/m$)



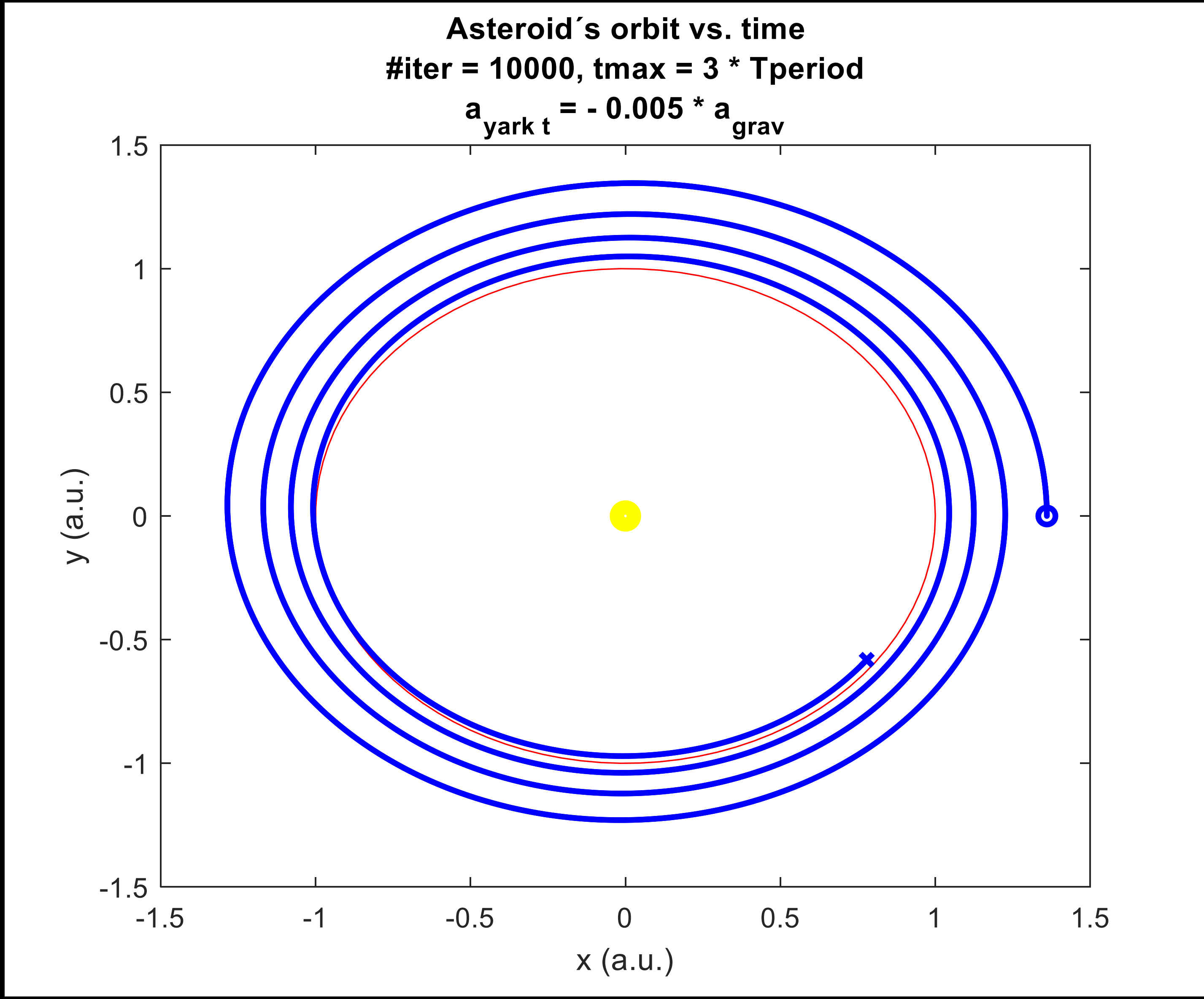
Orbital simulation – energy conservation





Orbital simulation w/ Yarkowsky

- Tangential Yarkowsky accel. is now added.
- Noticeable orbit spiral in/out depending on the sign of the acceleration.
- Acceleration value exaggerated on purpose.



Orbital simulation w/ Yarkowsky

- Measured semi-major axis drift for Bennu:

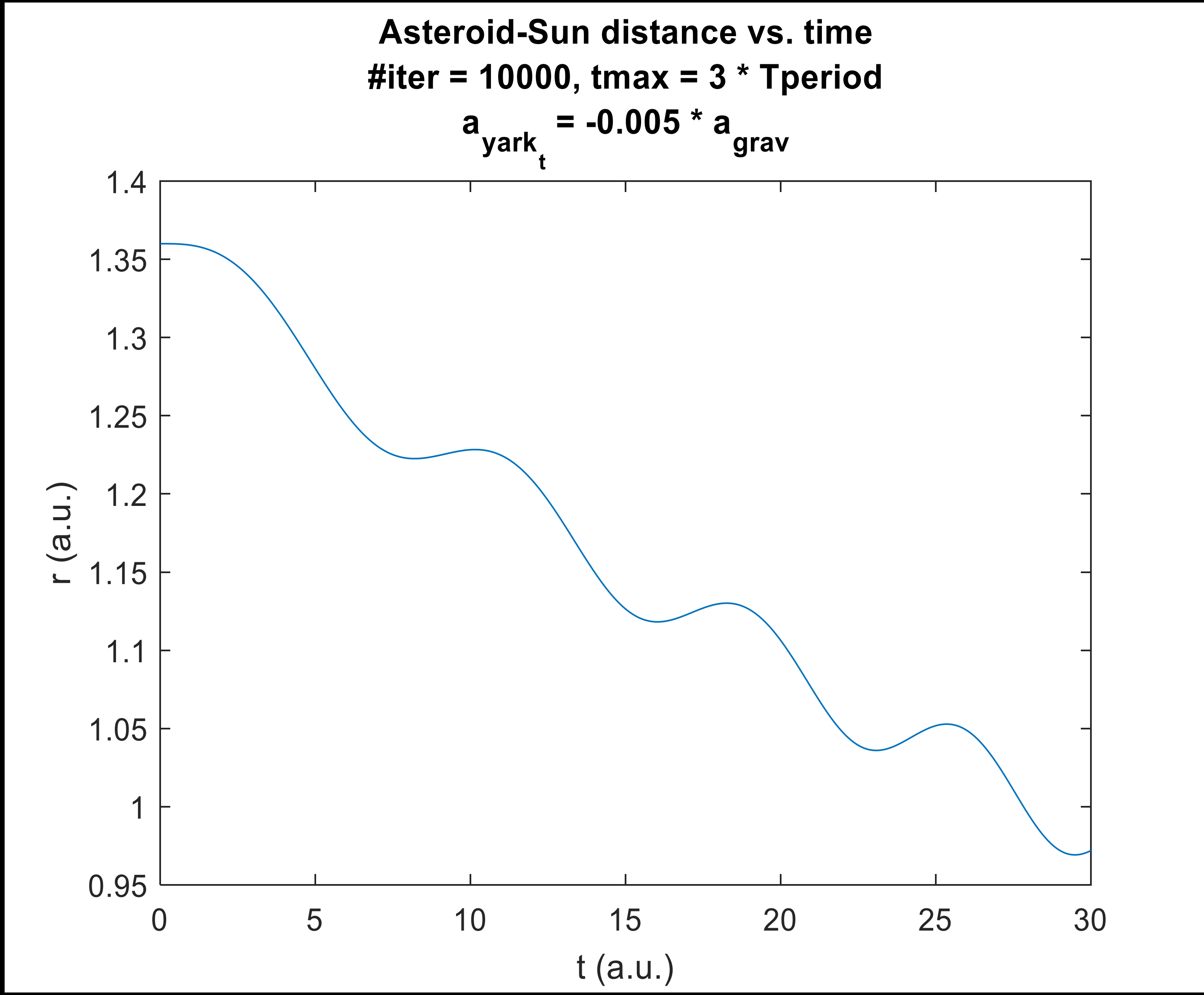
  $(-18.95 \pm 0.10) \times 10^{-4}$ AU/Myr

  ~ 0.25 Earth radii per Myr

- Theoretical calculation using available data:

  $(-16.5) \times 10^{-4}$ AU/Myr

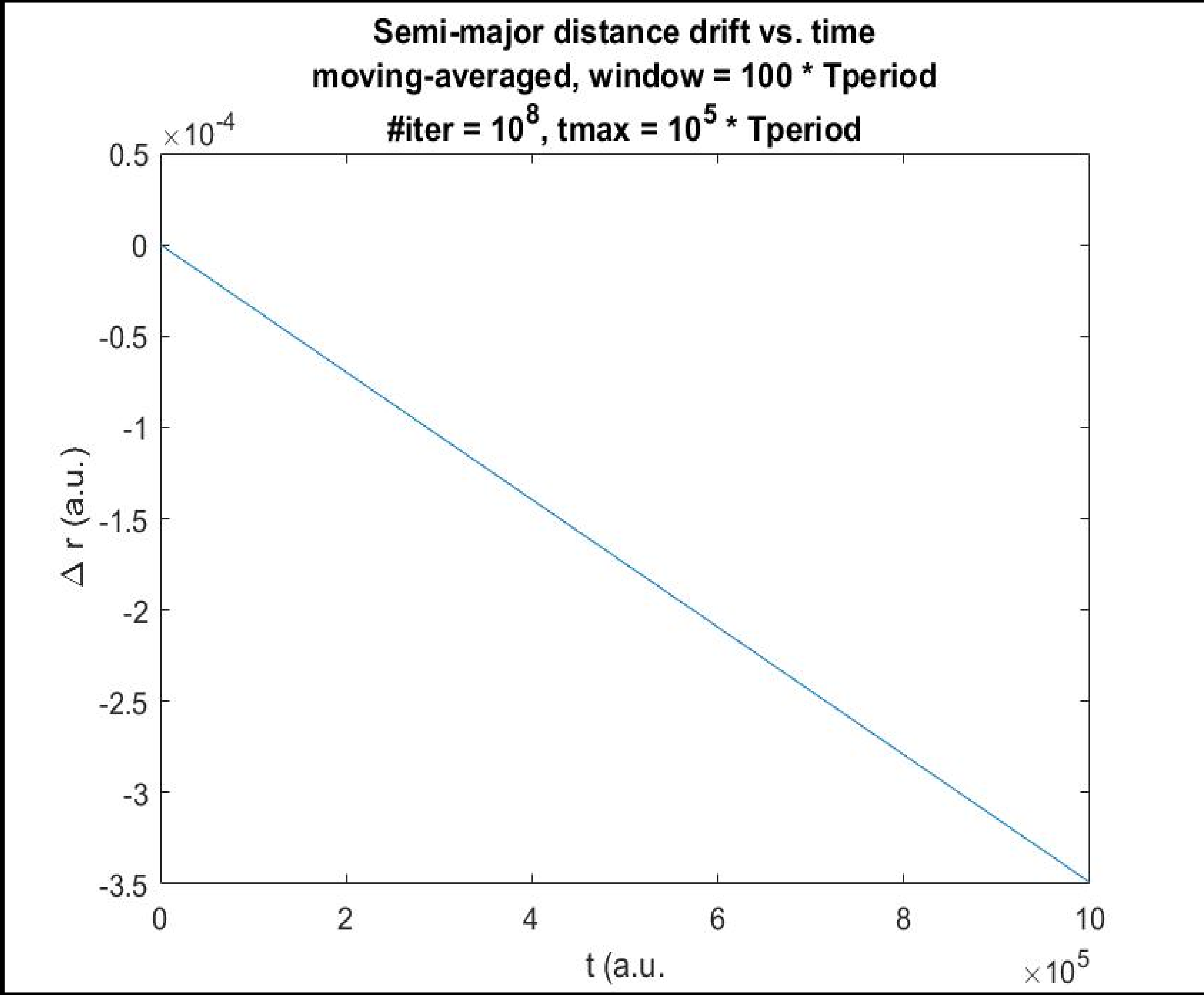
  [Vokrouhlicky 2015]



Orbital simulation w/ Yarkowsky

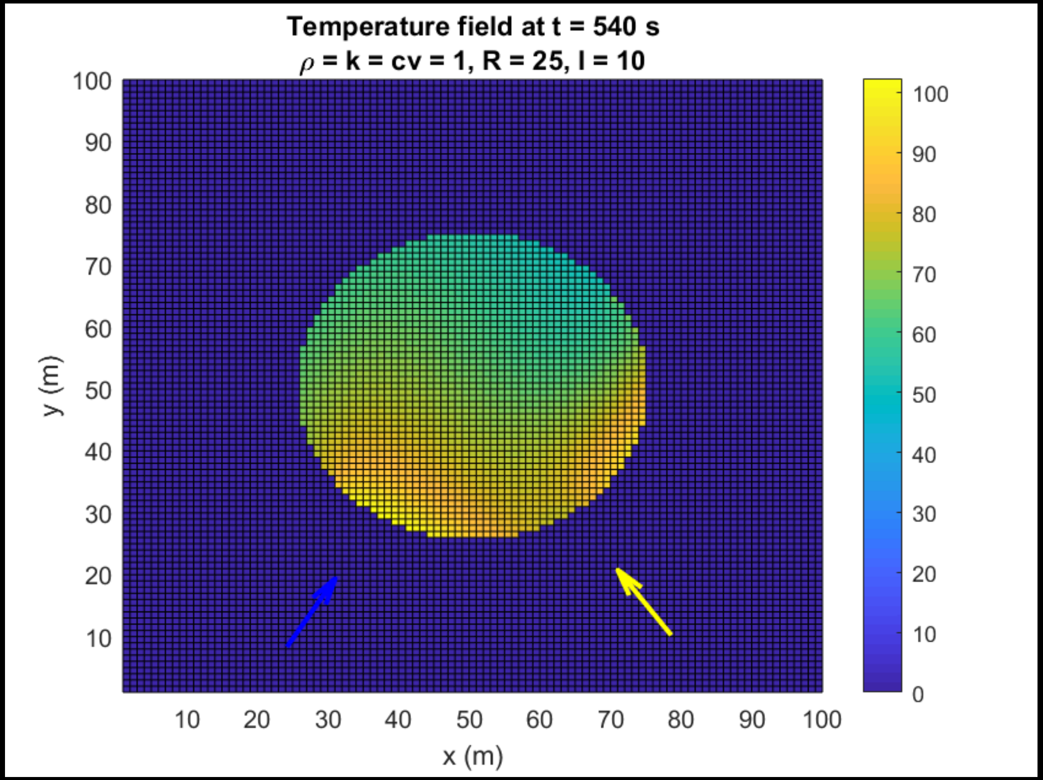
- Measured semi-major axis drift for Bennu:
  $(-18.95 \pm 0.10) \times 10^{-4}$ AU/Myr
  ~ 0.25 Earth radii per Myr
- Theoretical calculation using available data:
  $(-16.5) \times 10^{-4}$ AU/Myr
- Simulated drift using theoretical value for Yarkowsky acceleration:
  $(-35.0) \times 10^{-4}$ AU/Myr

[Vokrouhlicky 2015]

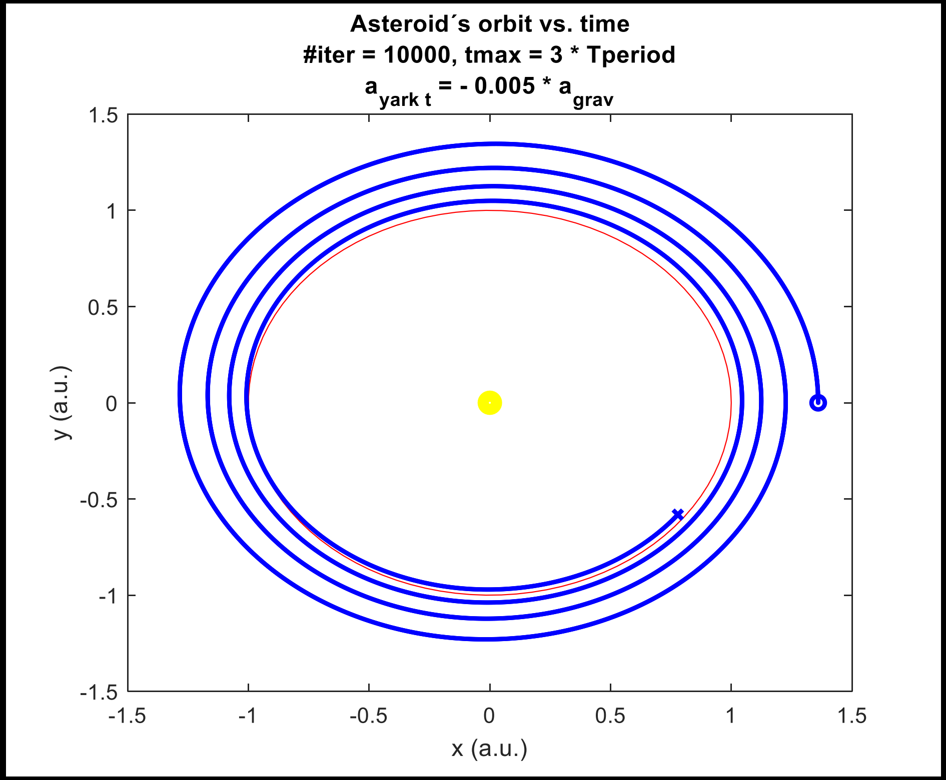


Summary

Simulation of Yarkowsky effect for a rotating asteroid works, but numerical (possibly algorithmical?) errors are still present.



Orbital simulation w/ Yarkowsky effect works, but obtained 2 higher drift than observed for Bennu(not great, not terrible)



Questions?

Image: NASA